



2D Graphics and Audio

Digital Media & Games
Derek Groen



Contents today

- 2D Graphics
 - Text
 - Images / Graphics
- Animation
 - Animation file formats
- Audio in Games

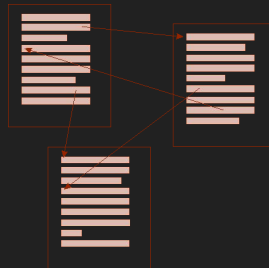
Text

Types of Text

- Unformatted text (Plain text)

Here's a rather boring and long block of text that is sure to make even the most curious of people extremely sleepy. In fact, I will be quite impressed if you read through the entire thing. I'll also have to ask if you really had nothing better to do than waste your time reading this pointless little exercise in wasting space. Ok, now that we've got that settled... What "that" you ask? That #1: You have no life. That #2: This example is really awful (or is it really good at depicting something really awful?). Whew, this is harder than I thought to write this much gibberish. I suppose that's a good thing, as you certainly wouldn't want to read something of this "quality" on a regular basis! If you have read this far, you have certainly earned my sympathy. Until next time...

- Formatted text (Rich text)



- Hypertext

This is a more interesting piece **of text showing a formatted piece of text** , including the use of **OTHER FONTS, SIZES**, the use of **bold**, *italics*, underlining AND **ALIGNMENTS**. **THERE MAY NOT BE** very much of interest in the actual content of the text, but it is more pleasing to **LOOK AT THAT PLAIN TEXT !**

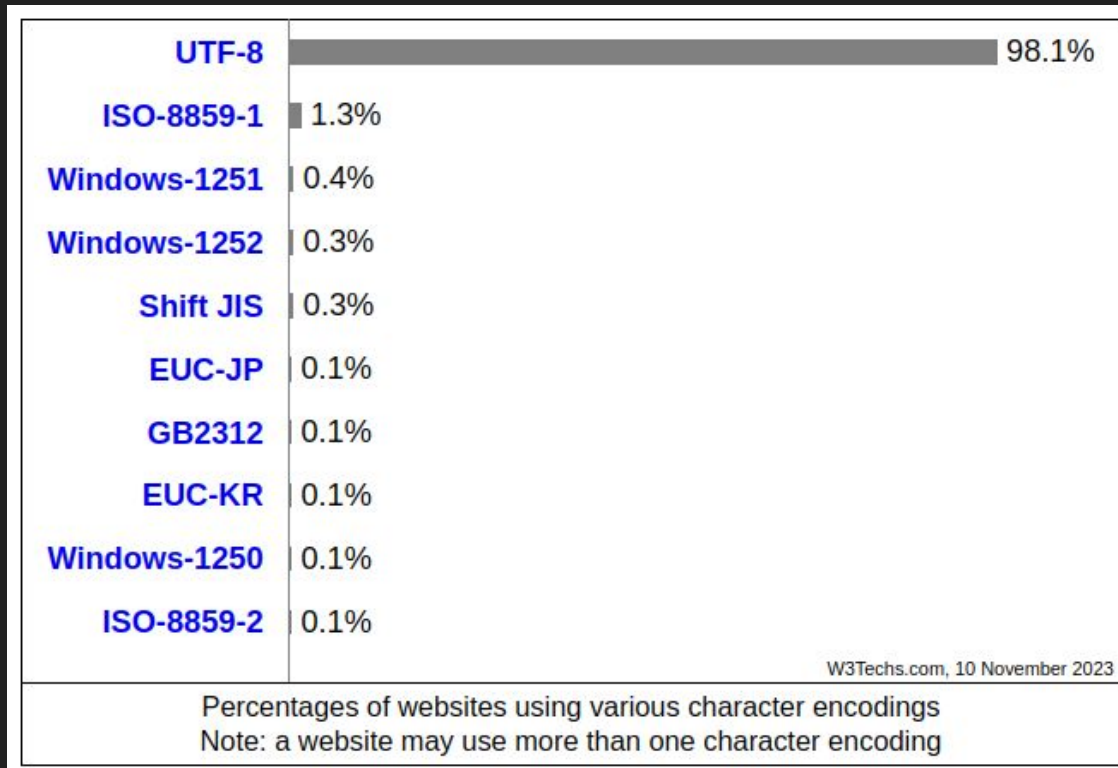
Plain / Unformatted Text

- **ASCII (≈1960)**

- 8-bit code (initially 128, later 256 characters to cater for several European languages)
- Common for programming.

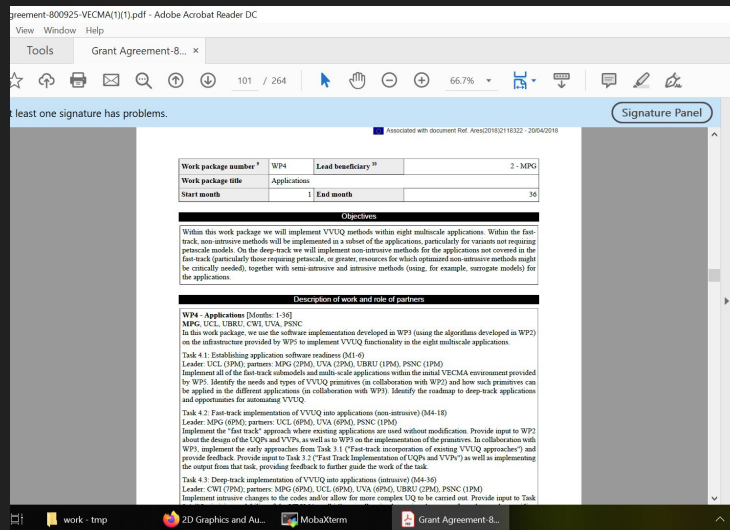
- **Unicode / UTF (1991)**

- Developed by Xerox Palo Alto Research Center (PARC) and Apple.
- Aim: to provide uniform character encoding.
- Common everywhere else.



https://w3techs.com/technologies/overview/character_encoding

Formatted Text



- Tables
- Columns
- Underline
- Equations

- Font types & sizes
- Headers
- Font effects

VECMA - 800925

stochastic variables; (iii) total sensitivity indices of uncertain parameters together with the stochastic variables. The shadow areas is one standard deviation of the estimators.

2.1.3 Stochastic Collocation

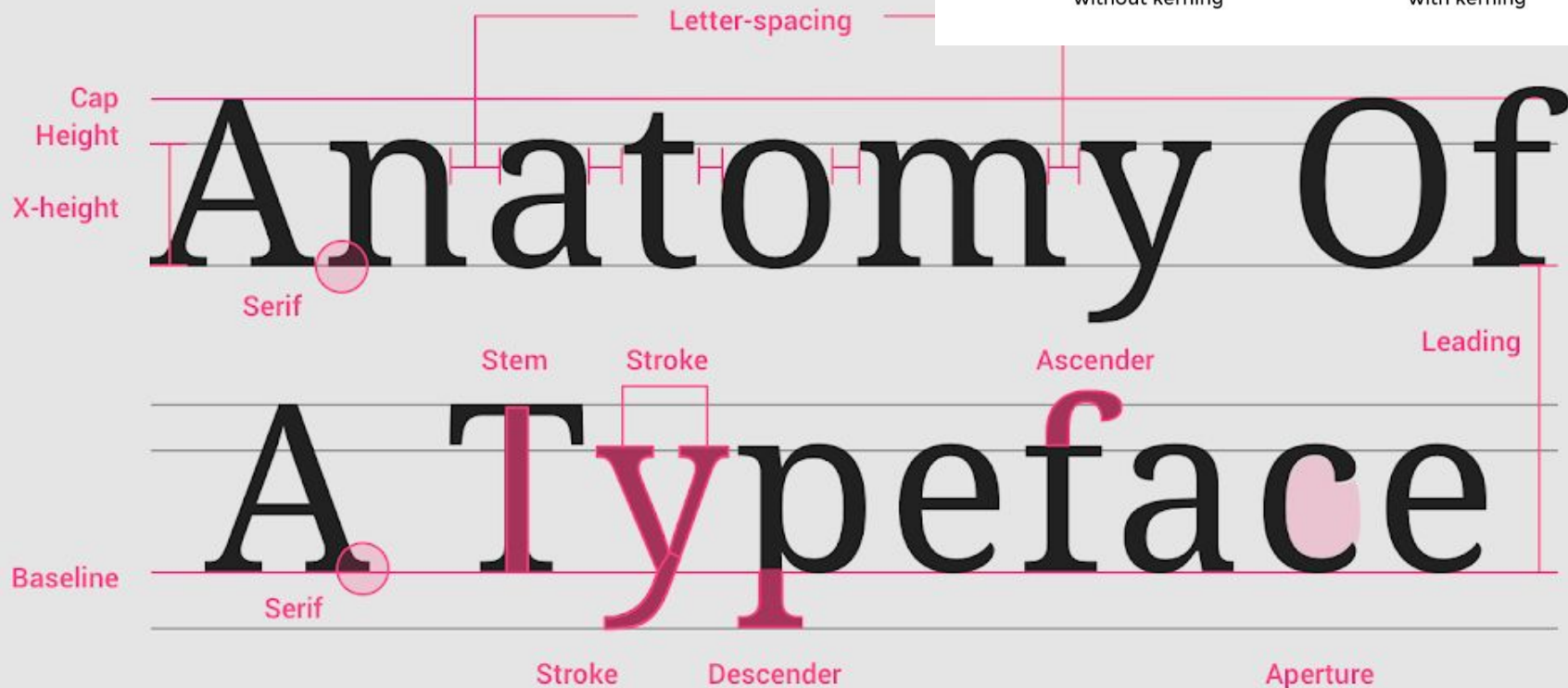
The SC method creates a polynomial approximation of a quantity of interest q in the (multi-dimensional) stochastic space ξ via the following expansion, written concisely as:

$$q(x, t; \xi) \approx \sum_{j=1}^{N_p} q_j(x, t) L(\xi_j)$$

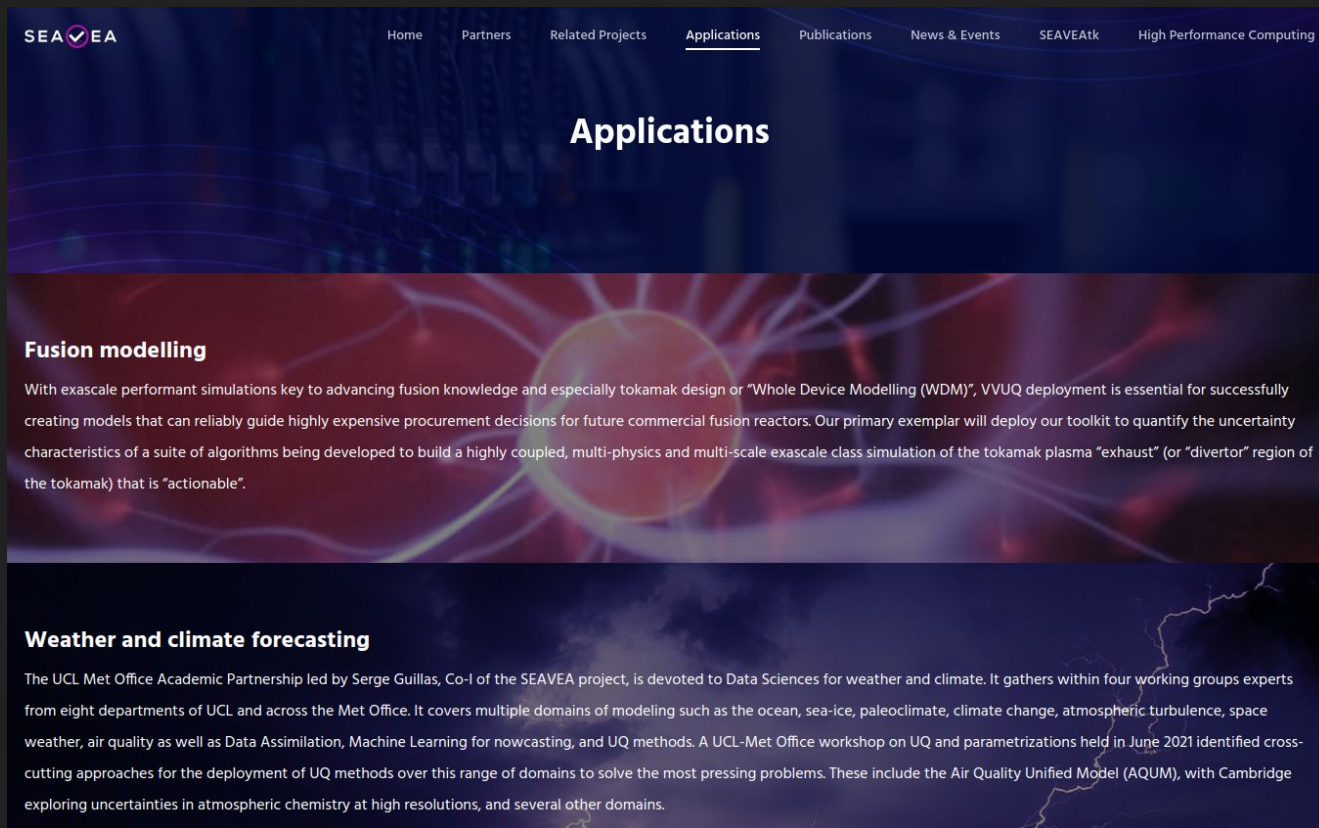
Here, q_j are the code samples which are computed on a structured multi-dimensional grid, and N_p is the total number of collocation points. The samples q_j are interpolated to an arbitrary point within the stochastic space by means of Lagrange interpolation polynomials $L(\xi)$. For interpolation in multiple dimensions, $L(\xi)$ is built as a tensor product of one-dimensional Lagrange polynomials. The SC method, as well as the PC method, are well-known and we refer to [19] for more details on these techniques.

Climate application

Text positioning and typefaces



Hypertext



SEAVEA

Home Partners Related Projects Applications Publications News & Events SEAVEAtk High Performance Computing

Applications

Fusion modelling

With exascale performant simulations key to advancing fusion knowledge and especially tokamak design or “Whole Device Modelling (WDM)”, VVUQ deployment is essential for successfully creating models that can reliably guide highly expensive procurement decisions for future commercial fusion reactors. Our primary exemplar will deploy our toolkit to quantify the uncertainty characteristics of a suite of algorithms being developed to build a highly coupled, multi-physics and multi-scale exascale class simulation of the tokamak plasma “exhaust” (or “divertor” region of the tokamak) that is “actionable”.

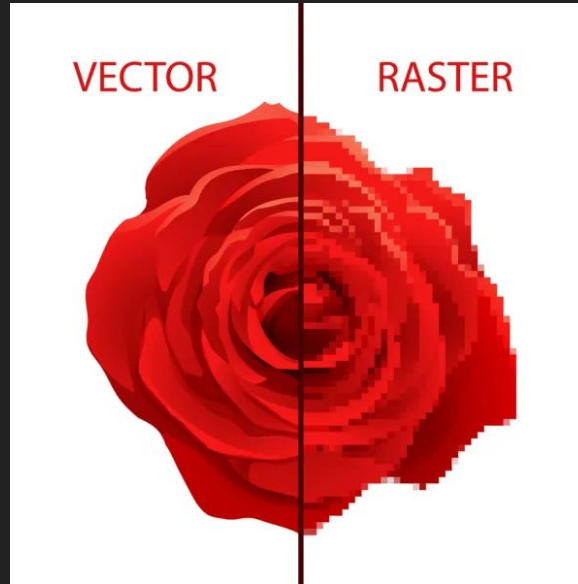
Weather and climate forecasting

The UCL Met Office Academic Partnership led by Serge Guillas, Co-I of the SEAVEA project, is devoted to Data Sciences for weather and climate. It gathers within four working groups experts from eight departments of UCL and across the Met Office. It covers multiple domains of modeling such as the ocean, sea-ice, paleoclimate, climate change, atmospheric turbulence, space weather, air quality as well as Data Assimilation, Machine Learning for nowcasting, and UQ methods. A UCL-Met Office workshop on UQ and parametrizations held in June 2021 identified cross-cutting approaches for the deployment of UQ methods over this range of domains to solve the most pressing problems. These include the Air Quality Unified Model (AQUM), with Cambridge exploring uncertainties in atmospheric chemistry at high resolutions, and several other domains.

2D graphics and images



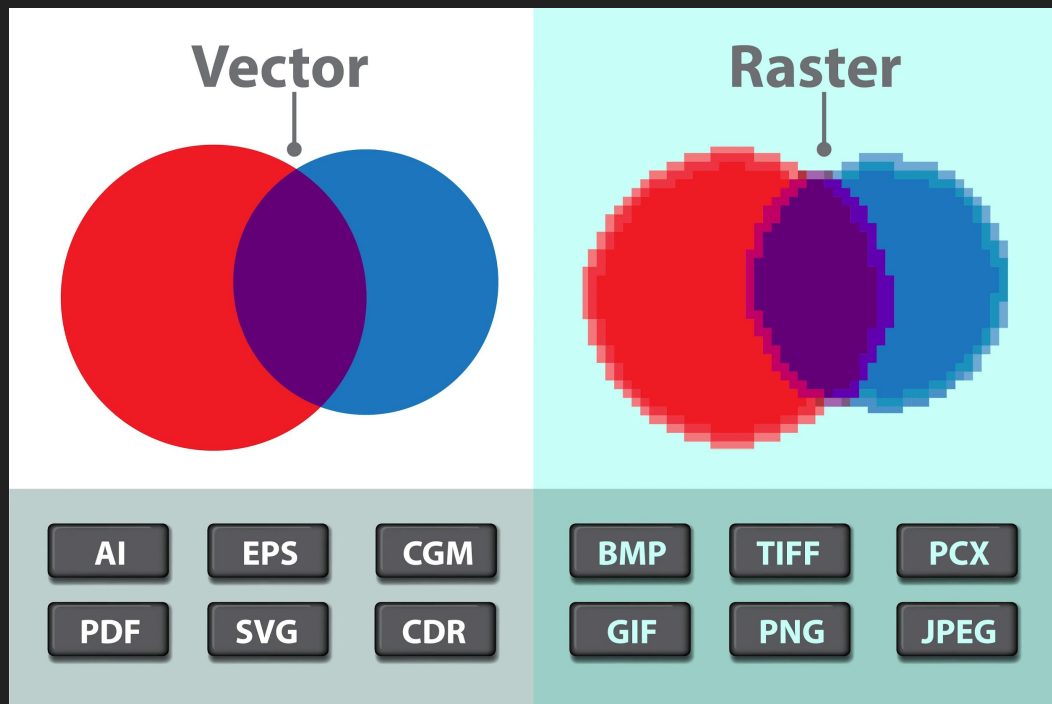
Vector vs. Raster images



Source: <https://tellyou rtale.com/blog/graphic-design/which-graphic-file-format-is-best-vector-and-raster-images/>

Graphics

- Vector:
 - Infinite zoom.
 - File size scales with the number of paths (objects).
- Raster:
 - Limited zoom.
 - File size scale with the resolution in pixels.



Anti-aliasing

Anti-aliasing is the application of subtle transitions in the pixels along the edges of images to minimize the jagged effect.

Anti-Aliased Bitmap Image:



Smooth Vector Image:



Monochrome Images

- Each pixel is stored as a **single bit** (0 or 1) so also referred to as binary image.
- Such an image is also called a 1-bit monochrome image since it contains no colour.
- Used for e.g. marquees and digital traffic indicators.

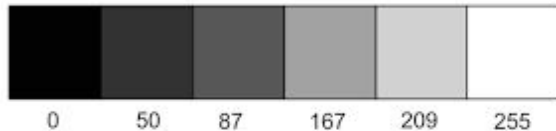


Grayscale Images

- **Each pixel has a gray-value between 0 and 255.** Each pixel is represented by a single byte; e.g., a dark pixel might have a value of 10, and a bright one might be 230.
- **Bitmap:** The two-dimensional array of pixel values that represents the graphics/image data.
- **Image resolution:** # of pixels in a digital image (higher = larger and more detailed).
 - HD = 1280 x 720 pixels
 - Full-HD = 1920 x 1080 pixels
 - Multiply the two numbers to get a resolution measured in Megapixels (e.g. for cameras).

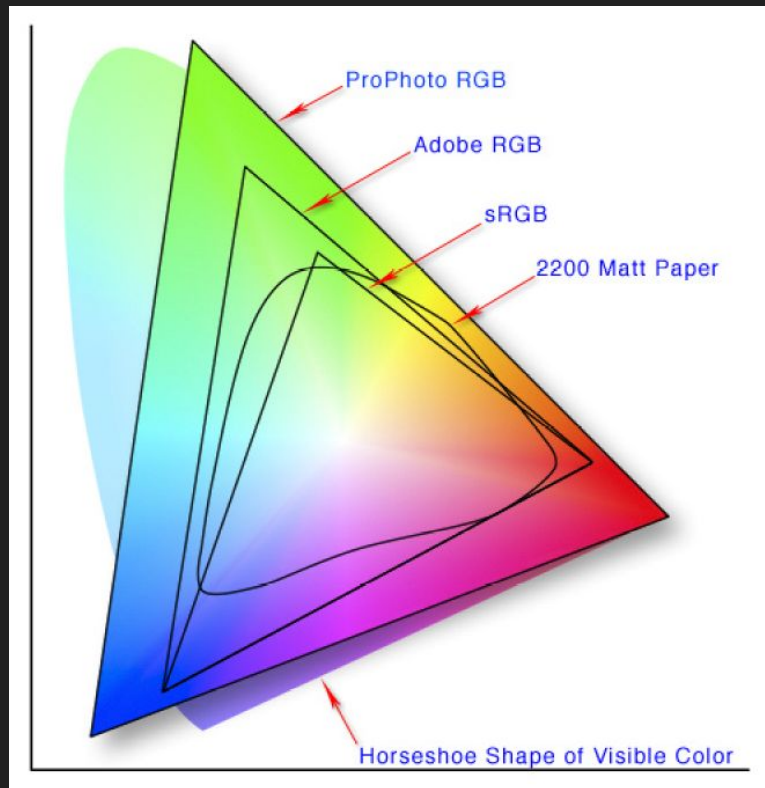
Grayscale Images

- Each pixel is usually stored as a byte (8 bits, value 0 to 255), so a 640 x 480 grayscale image requires ~300 kB of storage.
- (640 x 480 = 307,200).



24-bit Colour Images

- One bit per color, three color ranges:
 - Red
 - Green
 - Blue
- We cannot display all colors on common computer screens, even if we had infinite bits per color (!)



RGB and CMYK

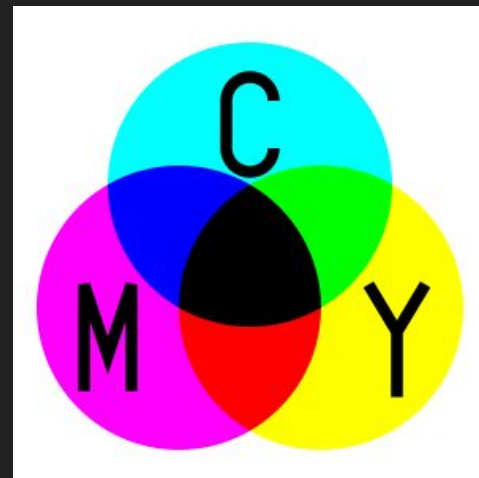
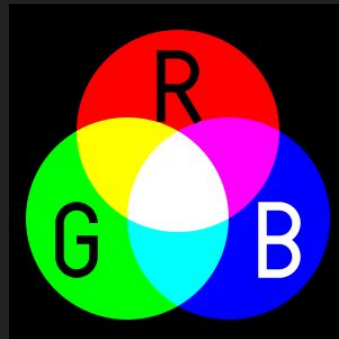
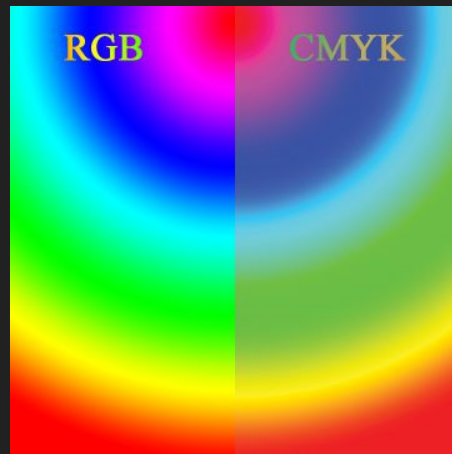
Screens have a black background, and the color space is additive:

- RGB model.

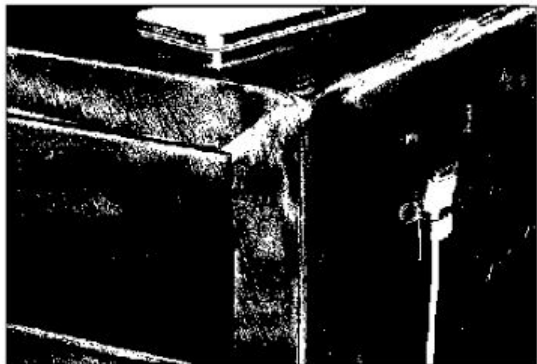
Printed media have a white background, and the color space therefore needs to be *subtractive*:

- Cyan-Magenta-Yellow-black model (CMYK).
- Why do we use black though?

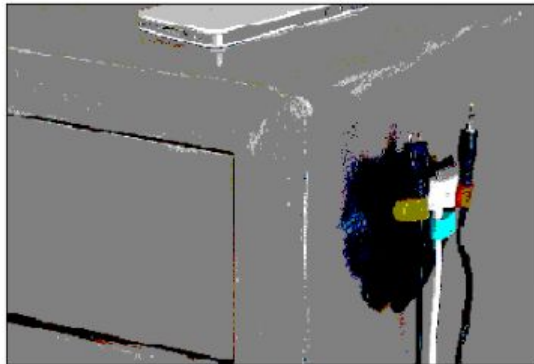
Some digital tools support CMYK to enable previews of graphics on printed media.



Color/Bit Depth Comparison



1 bit (2 colors)



4 bits (16 colors)



8 bits (256 colors)

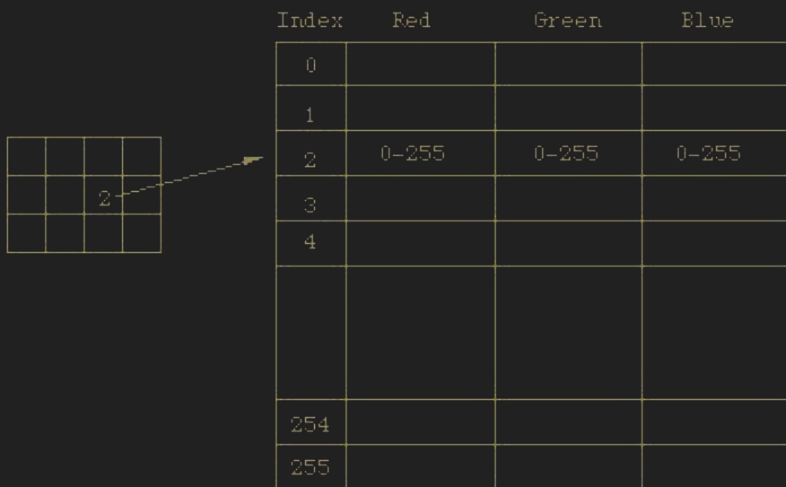


24 bits "true color" (16,777,216 colors)

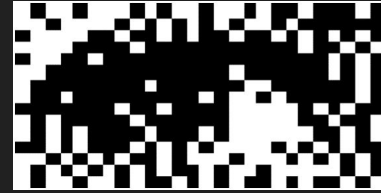
Indexed color

Each pixel has one byte associated with it, which is an index into a table of colours, called a palette or colour table.

Common in old consoles (e.g., NES used a 16-color palette based on a 256-color space), and used for animated gifs.



Dithering



Dithering takes advantage of the human eye's tendency to "mix" two colours in close proximity to one another.

left: grayscale image, middle: dithered version, right: detail of dithered version.

Image Data Types

- The most common color depths today are 24-bit colour and 8-bit colour.
- Some formats are restricted to particular hardware / operating system platforms, while others are “cross-platform” formats.
- Many use compression to reduce storage size, can be classified as **lossless** or **lossy**.
- Many tools enable format conversion (e.g., Imagemagick “convert” command on Linux).

24-bit Colour Images

- In a colour 24-bit image, each pixel is represented by three bytes, usually representing RGB.
 - This format supports $256 \times 256 \times 256$ possible combined colours, or a total of 16,777,216 possible colours.
 - However, the storage size is three times larger than that of a 8-bit image.
- **An important point:** many 24-bit colour images are actually stored as 32-bit images, with the extra byte of data for each pixel used to store an *alpha* value representing special effect information (e.g., transparency).



Raster images

Pixel-based graphics
Resolution dependent
Photos & web graphics

JPG

Web & print
photos and
quick previews

GIF

Animation &
transparency in
limited colors

PNG

Transparency
with millions
of colors

TIFF

High quality
print graphics
and scans

RAW

Unprocessed
data from
digital cameras

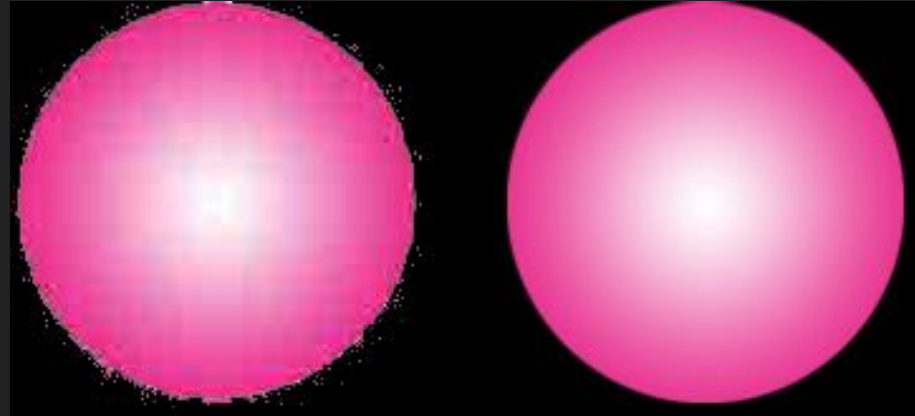
PSD

Layered Adobe
Photoshop
design files

Extension	Name	Notes
.jpg	Joint Photographic Experts Group	Lossy compression format well suited for photographic images.
.gif	Graphics Interchange Format	8-bit indexed bitmap format, is superseded by PNG on all accounts but animation.
.png	Portable Network Graphics	Lossless compression image, supporting 16-bit sample depth, and Alpha channel.
.tiff, .tif	Tagged Image File Format	Widely supported by scanning, faxing, word processing, optical character recognition, image manipulation, desktop publishing, and page-layout application.
.raw	Raw image file	Direct memory dump from a digital camera, contains the direct imprint from the imaging sensor, before layer interpolation and other colour corrections.
.psd	Photoshop Document	Native format of Adobe Photoshop, allows layers and other structural elements.

JPEG

- **JPEG:** Joint Photographic Experts Group.
 - JPEG uses discrete cosine transforms to provide lossy data compression, discarding small high-frequency patterns.
 - JPEG allows the user to set a desired level of quality, or compression ratio (input divided by output).
 - Poorly compressed JPEGs often feature “blocky” artifacts.



GIF

- Uses 8-bit color palettes, selected from a 24-bit color space.
 - Animated gifs have 1 palette per frame (!)
- Uses Lempel-Ziv-Welch lossless data compression, which optimizes for repeating patterns.
- GIF actually comes in three flavors:
 1. **GIF87a**: The original specification.
 2. **GIF89a**: Supports e.g. simple animation, provides simple control over delay time, and a transparency index.
 3. Netscape Application Block (NAB): supports (infinite) loops since Netscape 2.0.



PNG

- **PNG: Portable Network Graphics**
- Meant to supersede the GIF standard and extends it in important ways.
- Special features of PNG files include:
 - Support for up to 48 bits of colour information.
 - Files may contain gamma-correction information for correct display of colour images, as well as alpha-channel information to control transparency.
 - The display progressively displays pixels in a 2-dimensional fashion by showing a few pixels at a time over seven passes through each 8x8 block of an image.

TIFF

- **TIFF: stands for Tagged Image File Format.**
 - The support for attachment of additional information (referred to as “tags”) provide extra flexibility.
 - TIFF can store many different types of image: 1-bit, grayscale, 8-bit colour, 24-bit RGB, etc.
 - TIFF was originally a lossless format but now a new JPEG tag allows one to opt for JPEG compression.
 - Common in the printing and publishing industry.





Vector images

Curve-based graphics
Resolution independent
Logos, icons, & type

PDF

Print files and
web-based
documents

EPS

Individual
vector design
elements

AI

Original Adobe
Illustrator
design files

SVG

Vector files
for web
publishing

Extension	Name	Notes
.pdf	Portable Document Format	Save files that cannot be modified but still need to be easily shared and printed.
EPS, EPSF	Encapsulated PostScript File Format	Facilitate the incorporation of illustrations into textual documents for printing and transfers PostScript artwork between applications
.ai	Adobe Illustrator Artwork file	Vector based graphics contained in a single page and the graphics retains their details at maximum zoom level.
.svg	Scalable Vector Graphics	Extensible Markup Language (XML) based vector image format for two-dimensional graphics with support for interactivity and animation.

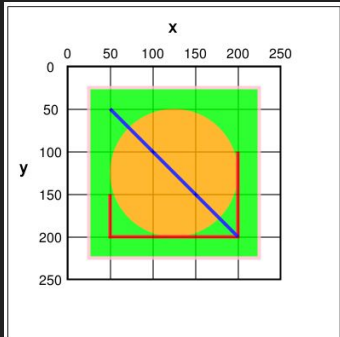
PS and PDF

- **Postscript** is an important language for typesetting, and many printers have a Postscript interpreter built into them.
- Vector-based picture language:
 - Postscript includes text as well as vector/structured graphics.
 - GL bit-mapped images can be included in output files.
 - Encapsulated Postscript files add some additional information for inclusion of Postscript files in another document.
 - Postscript page description language itself does not provide compression; in fact, Postscript files are just stored as ASCII.

SVG



- Developed in 1999 by the W3C.
- Supports vector graphics shapes, text and bitmap placement.
- Objects can be styled, transformed, grouped and composited.
- Support for interactive drawings and animations (although not applied on all platforms).



```
<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<!DOCTYPE svg PUBLIC "-//W3C//DTD SVG 1.1//EN" "http://www.w3.org/Graphics/SVG/1.1/DTD/svg11.dtd">
<svg width="391" height="391" viewBox="-70.5 -70.5 391 391" xmlns="http://www.w3.org/2000/svg" xmlns:xlink="http://www.w3.org/1999/xlink">
  <rect fill="#fff" stroke="#000" x="-70" y="-70" width="390" height="390"/>
  <g opacity="0.8">
    <rect x="25" y="25" width="200" height="200" fill="lime" stroke-width="4" stroke="pink" />
    <circle cx="125" cy="125" r="75" fill="orange" />
    <polyline points="50,150 50,200 200,200 200,100" stroke="red" stroke-width="4" fill="none" />
    <line x1="50" y1="50" x2="200" y2="200" stroke="blue" stroke-width="4" />
  </g>
</svg>
```


Animations

Animation

- Image designed to change over a period of time.
- Animation techniques are common just about anywhere (e.g., games, films, websites).
- 2D and 3D animation (creates illusion of depth)



Creating an animation

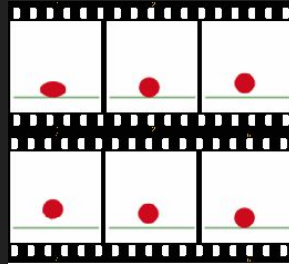
To create an animation:

- Organise the execution in a series of logical steps
- Choose an animation tool best suited for the task
- Build and tweak the sequence
- Postprocess the completed animation.

Animation Types

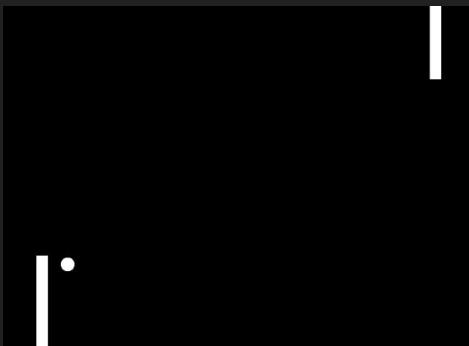
- **Hide and show animation**

- all objects created
- selected number shown
- timing of when objects are visible creates the illusion of animation



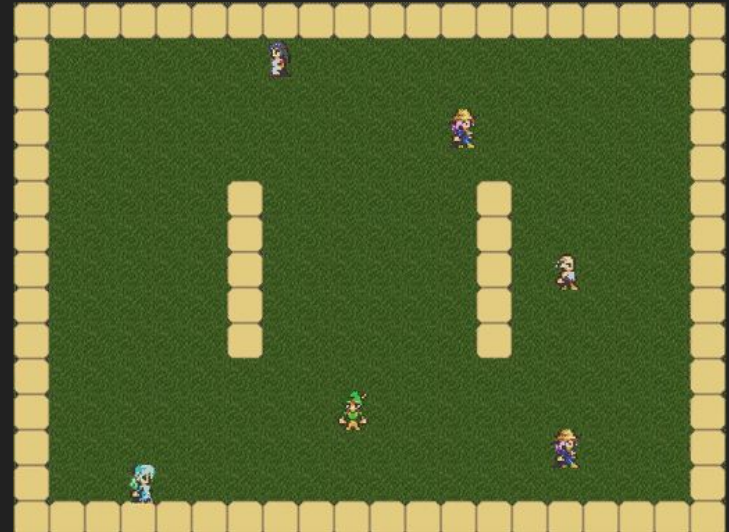
Animation Types: moving objects

- Objects are placed on the screen and controlled by a software program which moves them around the screen as required.



Animation types: image replacement

- Entities are animated by changing in between frames in a sprite set.
- Different frames may be rendered based on e.g., time intervals or the orientation/facing of the entity.



Animation Types: Zoom / Resize

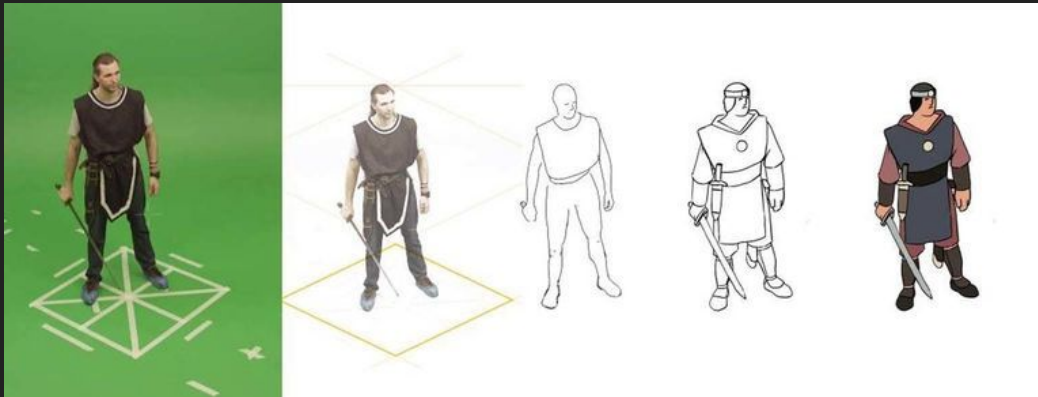
- The boundaries of objects on the screen are changed under program control.
- By resizing a series of objects on a screen one can create the impression of animation.



(okay, this one rotates too but you get the point ;))

Animation Types: Rotoscoping

- Take photographs or a video, extract essential elements and then draw on top of it.
- Widely used in traditional animation (e.g. Snow White 1951), but still in use today.
- Motion capture is an (advanced form) of rotoscoping.



<https://classpert.com/blog/all-about-roto-scoping>

Animation Types: Colour cycle animation

- Change values of colour components.
- Can e.g. create the illusion of flows.
- Used in old school computer games.



<https://imgur.com/r/pixelart/zfmq5TZ>

Keyframing and inbetweening

A **keyframe** in animation and filmmaking is a drawing that defines the starting and ending points of any smooth transition.

Programs can then interpolate the frames in between.

This is called *inbetweening* or *tweening*.

One can tween based on shape, color, or location.

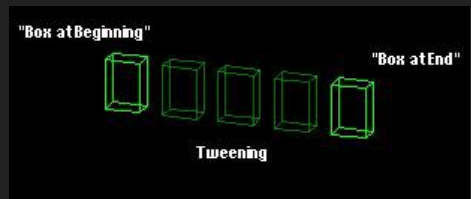
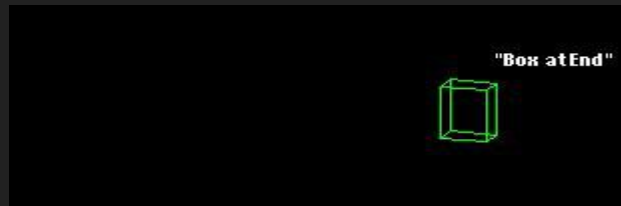


Image examples in the coursework

- Sprites:

- Figures within a game that usually have a transparent outline and can be animated to simulate movement.



- Backgrounds:

- Images that are placed behind the game objects to show the game environment.



Animation file formats

- “Silent” animation formats (like animated gifs)
- Full motion video formats
- Streaming formats

.Gif and .gifv

- GIF files can be animated...
- But compression is poor, and file sizes can get very large.
- .gifv is now widely used, but is not a video file in the classical sense.
 - Actually a HTML file with a HTML5 video tag to a proper movie format.
 - File size can be around 95% smaller than normal .gif files.
 - Used by imgur.com. Sites like gfycat.com do similar conversions.

APNG

- Animated version of PNG files, similar to animated GIFs.
- First image is written as a PNG stream, allowing the first frame to be visualized on non-supporting platforms.
- Established in 2008.
- 24-bit color + 8-bit transparency.
- Better compression performance than animated gifs or webp files.



WEBP

- Animation format that supports both lossless and lossy compression.
- Originally developed by Google, widely used today, but not supported by Google Slides or Unity (!)
- Doesn't clearly outperform JPEG for still images, and doesn't clearly outperform APNG for animations...

Game Audio

Based on slides by Karen Collins

<https://www.slideshare.net/collinsk/video-game-music-overview>



Game audio includes

- Dialogue / speech sounds.
- Sound effects.
- Music / score.
- Ambience.
- Interface sounds.



On audio

What I will cover:

- Use of audio specifically in games.
 - Audio types.
 - Main purposes.
 - Common problems.

What I will not cover:

- Audio formats.
- Audio filtering.
- Audio compression.

Good resource on basic aspects of sound:

<https://m2.material.io/design/sound/sound-attributes.html#type-properties>



Types of game audio

- Static
 - Ongoing throughout the game, e.g. background music.
- Interactive
 - Triggered by specific actions from the player.
- Adaptive
 - Triggered by changes in the gameplay environment, but not directly by player actions.
- Dynamic
 - Refers to both interactive and adaptive audio.



Functions of game sound

- Commercial functions.
- Reinforce game structure.
- Immersion / suspension of disbelief.
- Semiotics functions.
- Narrative functions.
- Emotional functions.
- Aesthetic functions.
- Kinetic functions.



Semiotic functions

- Sonic symbols / leitmotifs: identify symbols, decrease steepness of learning curve.
- Preparatory function.
- Convey meaning, emotion.
- Add slant or bias.



Narrative functions

- Advance plot.
- Locate player in storyline.
- Foreshadow.
- Access character's thoughts / reveal character's details.
- Reveal goals.



Emotional, aesthetic & kinetic functions

- Emotional / physiological: manipulate player.
- Create moods.
- Create intimacy.
- Make a game “cool”.
- Indicate genre/style of a game (e.g., use of a western tune).
- Music as motivating factor for movement.
- Edutainment for toddlers etc.



Common audio problems in games

- Mixing: too many sounds, timing errors, no “post-production”.
- Poor sampling/compression
 - Caused by space limitations.
- Transitions from cue to cue
 - Abrupt jumps between sounds, bad or no transitions.
 - Solved e.g. with cross-fades.
- Listener fatigue
 - Use silence(!), avoid infinite loops, variability.

Thank you!

Raster image File Extensions

Extension	Name	Notes
.gif	Graphics Interchange Format	8-bit indexed bitmap format, is superseded by PNG on all accounts but animation
.jpg	Joint Photographic Experts Group	Lossy compression format well suited for photographic images
.png	Portable Network Graphics	Lossless compression image, supporting 16bit sample depth, and Alpha channel
.tiff, .tif	Tagged Image File Format	TIFF is widely supported by scanning, faxing, word processing, optical character recognition, image manipulation, desktop publishing, and page-layout application
.psd	Photoshop Document	Native format of Adobe Photoshop, allows layers and other structural elements
.raw, .raw	Raw image file	Direct memory dump from a digital camera, contains the direct imprint from the imaging sensor, before layer interpolation and other colour corrections.

Animation Files

1. **FLC** is an animation or moving picture file format; it was originally created by Animation Pro. Another format, FLI, is similar to FLC.
2. **GL** produces somewhat better quality moving pictures. GL animations can also usually handle larger file sizes.
3. Many older formats: such as **DL or Amiga IFF files**, **Apple Quicktime files**, as well as animated **GIF89** files.

EXIF

- **EXIF** (Exchange Image File) is an image format for digital cameras:
 - Compressed EXIF files use the baseline JPEG format.
 - A variety of tags (many more than in TIFF) are available to facilitate higher quality printing, since information about the camera and picture-taking conditions (flash, exposure, light source, white balance, type of scene, etc.) can be stored and used by printers for possible colour correction algorithms.
 - The EXIF standard also includes specification of file format for audio that accompanies digital images.

Some Other Image Formats

- ▶ **Microsoft Windows: WMF:** the native vector file format for the Microsoft Windows operating environment:
 - ▶ Consist of a collection of GDI (Graphics Device Interface) function calls, also native to the Windows environment.
 - ▶ When a WMF file is “played” (typically using the Windows PlayMetaFile() function) the described graphics is rendered.
 - ▶ WMF files are ostensibly device-independent and are unlimited in size.
- ▶ **Microsoft Windows: BMP:** the major system standard graphics file format for Microsoft Windows, used in Microsoft Paint and other programs. Many sub-variants within the BMP standard.
- ▶ **Macintosh: PAINT and PICT:**
 - ▶ PAINT was originally used in the MacPaint program, initially only for 1-bit monochrome images.
 - ▶ PICT format is used in MacDraw (a vector-based drawing program) for storing structured graphics.
- ▶ **X-windows: PPM:** the graphics format for the X Window system. PPM supports 24-bit colour bitmaps and can be manipulated using many public domain graphic